HK IMO Training 2021-2022
Problem Set 11
For 18 December 2021

Problem 41. Let $n \geqslant 3$ be an integer. Let $a_1, a_2, \ldots, a_n$ be n pairwise distinct integers. Find the smallest possible value of

$$(a_1 - a_2)^2 + (a_2 - a_3)^2 + \cdots + (a_{n-1} - a_n)^2 + (a_n - a_1)^2.$$

Problem 42. Let n be a positive integer. There are $4n$ coins lying in a row, including $2n$ one-dollar coins and $2n$ two-dollar coins. In each move, we can choose k consecutive one-dollar coins and choose k consecutive two-dollar coins, and swap these two groups of coins, where k can be any positive integer that can vary in different moves. Find the minimum number of moves needed so that we can always reach the state in which the first $2n$ coins are one-dollar coins, regardless of the initial positions of the coins.

Problem 43. Let $a_n = 2^{n^2+1} - 3^n$ for every positive integer n.

(a) Prove that there are infinitely many prime numbers dividing some of the a_n's.

(b) Prove that there are infinitely many prime numbers not dividing any of the a_n's.

Problem 44. Let Γ be the circumcircle of an acute $\triangle ABC$. Let M be the midpoint of BC, and let D be the midpoint of the minor arc BC of Γ. Let Ω be the circle with centre D that passes through M. The two tangents from A to Ω meet the side BC at E and F respectively, with E lying between B and F. A circle ω_1 is tangent to the segments AE and CE, and is also tangent to Γ at P. A circle ω_2 is tangent to the segments AF and BF, and is also tangent to Γ at Q. Let X be the intersection point of EP and FQ. Prove that $\angle EAX = \angle MAF$.